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**ARTIFICIAL LIGHT AT NIGHT AND BAT ACTIVITY**  
ALONG A DISTANCE GRADIENT FROM THE COLTER BAY PARKING LOT  
Grand Teton National Park, 2022  
Nigel K. Anderson · American Museum of Natural History

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**OVERVIEW**

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Study question: does light color (red vs. white) affect bat activity, and does the answer depend on intensity or distance from the Colter Bay parking lot?

Design: 14 acoustic monitoring stations along a distance gradient (<0.52 km to >2.1 km from Colter Bay parking lot), red vs. white light crossed at five intensities (10, 30, 50, 70, 100%). Survey season: 2022. Six focal species: Epfu, Laci, Lano, Myev, Mylu, Myvo.

Six species with sufficient data are further characterized by four continuous wing-morphology traits (ear height, forearm length, wing loading, aspect ratio). Second, the same light treatments were run past moth (Lepidoptera) data at all 14 stations over the same season, identified to species, allowing a direct bat-vs-moth comparison of the light color, intensity, and distance effects (Sections 9-10).

Additional supporting figures are collected in Supplemental Materials at the end.

Overall main results: Bats were most active under red light, but only when it was dim to moderately bright. At the highest brightness, red and white lighting made no difference to the bats. Bats were more active further from the bright Colter Bay parking lot, but this pattern faded out under high brightness. The smaller Myotis species were the ones that showed up further away from the parking lot while the larger bats were more prevalent at Colter Bay.

Moths showed the opposite color preference. They preferred white light at high light intensity. However, we did find that moths were less prevalent the closer the sites were to Colter Bay.

We of course have more results but I think these are the main findings. We find that moon phase and seasonality impact bat detections (differently for red and white). We also find that some species are being impacted by sky brightness at the different sites and also the moon phase.

## **PRIMARY MODEL**

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Family: nbinom2 ( log )  
Formula: detections ~ color \* intensity + intensity \* dist\_km +  
mean\_phase + pct\_nonforest + brightness\_dark +

```

jd_c + I(jd_c^2) + (1 | site)
Zero infl: ~1
Data:      data_env

```

| AIC     | BIC     | logLik   | -2*log(L) | df.resid |
|---------|---------|----------|-----------|----------|
| 24256.7 | 24398.1 | -12105.4 | 24210.7   | 3434     |

Random effects:

| Groups | Name        | Variance | Std.Dev. |
|--------|-------------|----------|----------|
| site   | (Intercept) | 0.3728   | 0.6106   |

Number of obs: 3457, groups: site, 14

Dispersion parameter for nbinom2 family (): 0.607

Conditional model:

|                      | Estimate   | Std.Err | z value | Pr(> z ) |     |
|----------------------|------------|---------|---------|----------|-----|
| (Intercept)          | 2.1920095  | 0.3045  | 7.199   | 6.05e-13 | *** |
| colorW               | 0.0499199  | 0.0931  | 0.536   | 0.5917   |     |
| intensity30          | 0.7155529  | 0.1476  | 4.849   | 1.24e-06 | *** |
| intensity50          | 0.4606876  | 0.1569  | 2.936   | 0.0033   | **  |
| intensity70          | 0.6229863  | 0.1099  | 5.671   | 1.42e-08 | *** |
| intensity100         | 0.2331022  | 0.1331  | 1.752   | 0.0798   | .   |
| dist_km              | 0.1375974  | 0.0675  | 2.040   | 0.0414   | *   |
| mean_phase           | 0.1992094  | 0.1171  | 1.701   | 0.0889   | .   |
| pct_nonforest        | 0.5326479  | 0.8559  | 0.622   | 0.5337   |     |
| brightness_dark      | -0.0262336 | 0.0376  | -0.698  | 0.4854   |     |
| jd_c                 | -0.0073183 | 0.0017  | -4.287  | 1.81e-05 | *** |
| I(jd_c^2)            | -0.0003733 | 0.0002  | -2.150  | 0.0315   | *   |
| colorW:intensity30   | -0.7703147 | 0.1751  | -4.399  | 1.09e-05 | *** |
| colorW:intensity50   | -0.5357506 | 0.1459  | -3.672  | 0.0002   | *** |
| colorW:intensity70   | -0.1550017 | 0.1366  | -1.135  | 0.2565   |     |
| colorW:intensity100  | 0.0172641  | 0.1353  | 0.128   | 0.8985   |     |
| intensity30:dist_km  | -0.0501091 | 0.0319  | -1.572  | 0.1159   |     |
| intensity50:dist_km  | -0.0364428 | 0.0294  | -1.239  | 0.2153   |     |
| intensity70:dist_km  | -0.1275481 | 0.0272  | -4.687  | 2.77e-06 | *** |
| intensity100:dist_km | -0.0897028 | 0.0269  | -3.335  | 0.0009   | *** |

Zero-inflation model:

| Estimate | Std.Err | z value | Pr(> z ) |
|----------|---------|---------|----------|
|----------|---------|---------|----------|

(Intercept)    -21.53    805.37    -0.027    0.979

Reference levels: color = R (red), intensity = 10%. Distance entered as continuous raw km (dist\_km). Two 2-way interactions: color × intensity, intensity × dist\_km.

#### POST-HOC: COLOR X INTENSITY

---

Estimated marginal means on the response scale (detections/night), averaged over distance. Holm-corrected pairwise contrasts.

=== Estimated marginal means: color at each intensity (response scale) ===

intensity = 10:

| color | response | SE   | df  | asympt.LCL | asympt.UCL |
|-------|----------|------|-----|------------|------------|
| R     | 12.7     | 2.29 | Inf | 8.93       | 18.1       |
| W     | 13.4     | 2.42 | Inf | 9.37       | 19.0       |

intensity = 30:

| color | response | SE   | df  | asympt.LCL | asympt.UCL |
|-------|----------|------|-----|------------|------------|
| R     | 23.2     | 4.64 | Inf | 15.70      | 34.3       |
| W     | 11.3     | 2.31 | Inf | 7.57       | 16.9       |

intensity = 50:

| color | response | SE   | df  | asympt.LCL | asympt.UCL |
|-------|----------|------|-----|------------|------------|
| R     | 18.6     | 3.90 | Inf | 12.30      | 28.0       |
| W     | 11.4     | 2.45 | Inf | 7.49       | 17.4       |

intensity = 70:

| color | response | SE   | df  | asympt.LCL | asympt.UCL |
|-------|----------|------|-----|------------|------------|
| R     | 17.8     | 3.21 | Inf | 12.49      | 25.4       |
| W     | 16.0     | 2.97 | Inf | 11.14      | 23.0       |

intensity = 100:

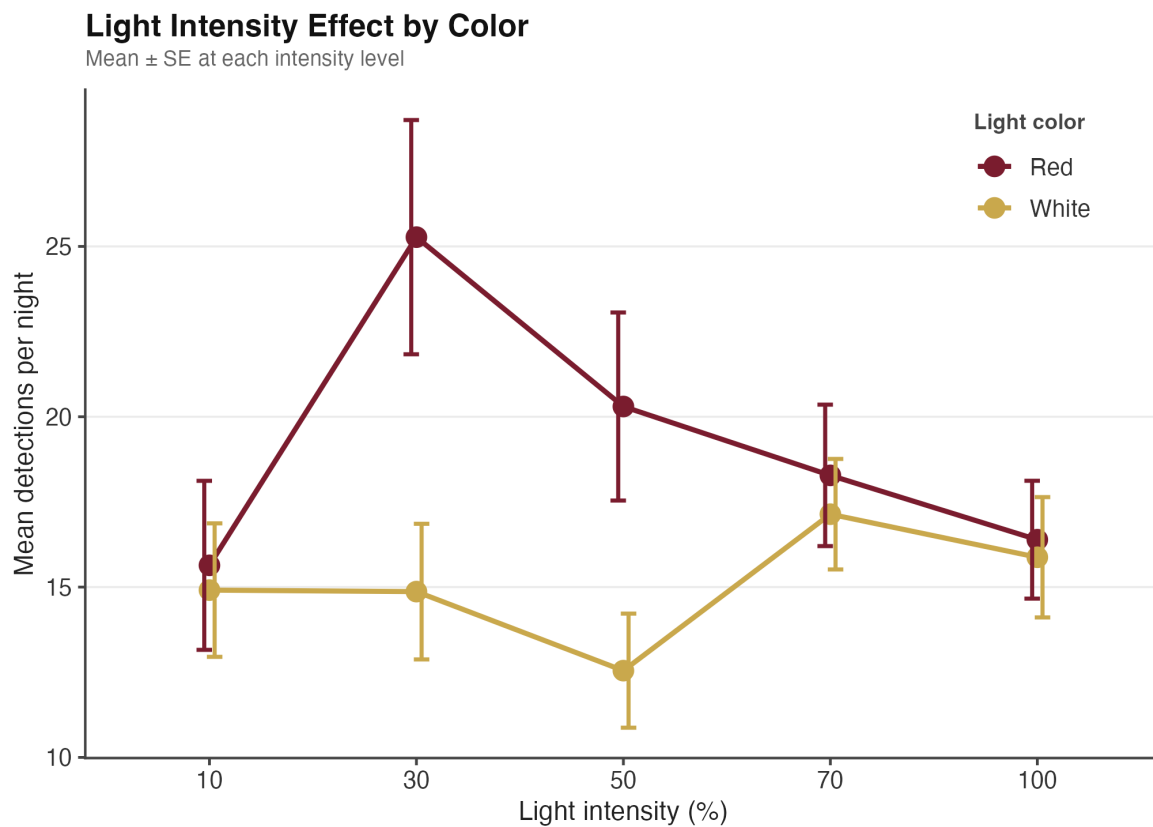
| color | response | SE   | df  | asympt.LCL | asympt.UCL |
|-------|----------|------|-----|------------|------------|
| R     | 13.1     | 2.48 | Inf | 9.05       | 19.0       |
| W     | 14.0     | 2.51 | Inf | 9.87       | 19.9       |

=== Red vs. White contrast at each intensity (Holm-corrected) ===

| intensity | ratio | SE    | df  | z.ratio | p.value     |
|-----------|-------|-------|-----|---------|-------------|
| 10        | 0.951 | 0.089 | Inf | -0.536  | 0.5917      |
| 30        | 2.055 | 0.299 | Inf | 4.946   | <0.0001 *** |
| 50        | 1.626 | 0.182 | Inf | 4.345   | <0.0001 *** |
| 70        | 1.111 | 0.109 | Inf | 1.067   | 0.2860      |
| 100       | 0.935 | 0.095 | Inf | -0.665  | 0.5062      |

Ratio = R / W (ratio > 1 = more detections under red)

At 30% intensity, red light produced 2.06x more detections than white ( $p < 0.0001$ ). At 50%, 1.63x ( $p < 0.0001$ ). At 70% and 100%, no significant difference. White and red converge above 50% intensity (Supplemental Figs. S1-S2).



**Figure 1:** Color  $\times$  intensity interaction: predicted detections from the GLMM on the response scale. Cardinal crimson = red light; gilded gold = white light.

## POST-HOC: INTENSITY WITHIN COLOR

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=== Intensity pairwise contrasts within Red (Holm-corrected) ===

| contrast | ratio | SE    | df  | z.ratio | p.value |     |
|----------|-------|-------|-----|---------|---------|-----|
| 10 / 30  | 0.547 | 0.075 | Inf | -4.406  | 0.0001  | *** |
| 10 / 50  | 0.685 | 0.098 | Inf | -2.654  | 0.0478  | *   |
| 10 / 70  | 0.714 | 0.066 | Inf | -3.619  | 0.0024  | **  |
| 10 / 100 | 0.969 | 0.113 | Inf | -0.272  | 1.0000  |     |
| 30 / 50  | 1.251 | 0.192 | Inf | 1.457   | 0.4350  |     |
| 30 / 70  | 1.305 | 0.182 | Inf | 1.912   | 0.2237  |     |
| 30 / 100 | 1.771 | 0.231 | Inf | 4.387   | 0.0001  | *** |
| 50 / 70  | 1.043 | 0.148 | Inf | 0.298   | 1.0000  |     |
| 50 / 100 | 1.415 | 0.171 | Inf | 2.880   | 0.0278  | *   |
| 70 / 100 | 1.356 | 0.163 | Inf | 2.539   | 0.0556  | .   |

=== Intensity pairwise contrasts within White (Holm-corrected) ===

| contrast | ratio | SE    | df  | z.ratio | p.value |
|----------|-------|-------|-----|---------|---------|
| 10 / 30  | 1.182 | 0.177 | Inf | 1.119   | 1.0000  |
| 10 / 50  | 1.170 | 0.184 | Inf | 0.999   | 1.0000  |
| 10 / 70  | 0.834 | 0.085 | Inf | -1.778  | 0.6033  |
| 10 / 100 | 0.952 | 0.096 | Inf | -0.484  | 1.0000  |

[all remaining contrasts: p >= 0.14]

Under red light, activity peaks at 30% (10/30 ratio = 0.547, i.e. 30% produces 1.83x more than 10%;  $p < 0.0001$ ) and collapses at 100% (30/100 ratio = 1.77;  $p < 0.0001$ ). 50% and 70% are intermediate and not distinguishable from 30%. White is flat across all intensities -- no pairwise contrast survives Holm correction (Supplemental Figs. S3-S4).

## POST-HOC: INTENSITY $\times$ DISTANCE

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Distance entered as continuous raw km. Interaction tests whether the effect of intensity on detections changes with distance from the Colter Bay parking lot.

=== Distance slope (km) by intensity level (emtrends) ===

| intensity | dist_km.trend | SE     | df  | asympt.LCL | asympt.UCL |   |
|-----------|---------------|--------|-----|------------|------------|---|
| 10        | 0.1376        | 0.0675 | Inf | 0.005      | 0.270      | * |
| 30        | 0.0875        | 0.0689 | Inf | -0.047     | 0.222      |   |
| 50        | 0.1012        | 0.0676 | Inf | -0.031     | 0.234      |   |
| 70        | 0.0100        | 0.0670 | Inf | -0.121     | 0.141      |   |
| 100       | 0.0479        | 0.0669 | Inf | -0.083     | 0.179      |   |

Slopes averaged over color levels. Positive = more detections farther from Colter Bay parking lot (recovery with distance). Only intensity 10 slope differs from zero (95% CI excludes 0).

=== Pairwise distance-slope contrasts (Holm-corrected) ===

| contrast          | estimate | SE    | df  | z.ratio | p.value |     |
|-------------------|----------|-------|-----|---------|---------|-----|
| intensity10 - 70  | 0.1275   | 0.027 | Inf | 4.687   | <0.0001 | *** |
| intensity10 - 100 | 0.0897   | 0.027 | Inf | 3.335   | 0.0077  | **  |
| intensity50 - 70  | 0.0911   | 0.028 | Inf | 3.222   | 0.0102  | *   |

[all other contrasts: p >= 0.085]

=== Intensity contrasts at ecologically meaningful distances ===

dist\_km = 0.3 km (Close):

| contrast          | ratio | SE    | df  | z.ratio | p.value |     |
|-------------------|-------|-------|-----|---------|---------|-----|
| intensity10 / 70  | 0.602 | 0.051 | Inf | -6.043  | <0.0001 | *** |
| intensity70 / 100 | 1.340 | 0.133 | Inf | 2.944   | 0.0292  | *   |

[all other contrasts: p >= 0.10]

dist\_km = 0.7 km (Medium):

| contrast          | ratio | SE    | df  | z.ratio | p.value |     |
|-------------------|-------|-------|-----|---------|---------|-----|
| intensity10 / 70  | 0.634 | 0.050 | Inf | -5.822  | <0.0001 | *** |
| intensity70 / 100 | 1.320 | 0.125 | Inf | 2.938   | 0.0297  | *   |

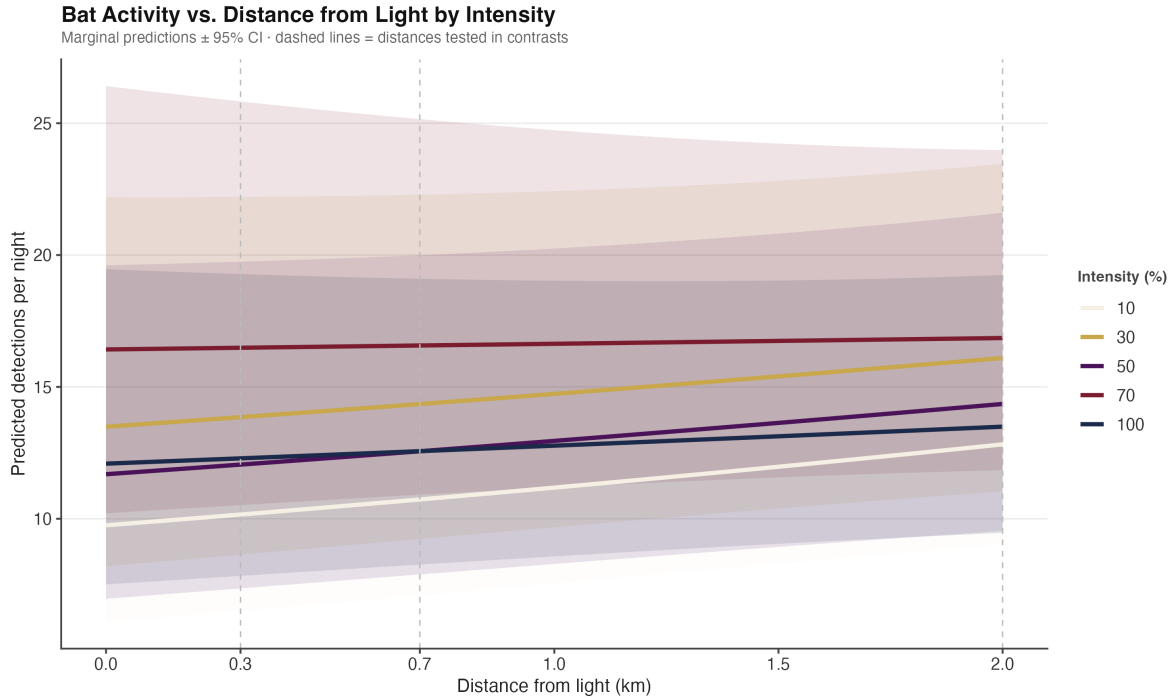
[all other contrasts: p >= 0.13]

dist\_km = 2.0 km (Far):

| contrast         | ratio | SE    | df  | z.ratio | p.value |     |
|------------------|-------|-------|-----|---------|---------|-----|
| intensity10 / 70 | 0.748 | 0.052 | Inf | -4.176  | 0.0003  | *** |

[all other contrasts: p >= 0.07]

The intensity  $\times$  distance interaction is driven by 70% and 100%: their distance slopes are significantly shallower than 10% (i.e. high intensities suppress bats uniformly across distance, not just close in). At 0.3 km, 70% produces 1.66x more detections than 10% (ratio = 0.602 inverted); this gap narrows but persists at 2.0 km (ratio = 0.748,  $p = 0.0003$ ). 70% and 100% converge only at far distances (additional distance-gradient views in Supplemental Figs. S5–S7).



**Figure 2:** Bat activity vs. distance from light by intensity. Marginal predictions  $\pm$  95% CI; dashed vertical lines mark the 0.3, 0.7, and 2.0 km distances tested in the pairwise contrasts above.

## COMMUNITY ANALYSIS

=== NMDS ===

Stress: 0.207 (k = 2, Bray-Curtis, trymax = 100)

=== PERMANOVA (adonis2) ===

|          | Df  | SumOfSqs | R2    | F     | Pr(>F)     |
|----------|-----|----------|-------|-------|------------|
| Model    | 9   | 3.8338   | 0.156 | 2.507 | <0.001 *** |
| Residual | 122 | 20.7295  | 0.844 |       |            |
| Total    | 131 | 24.5633  | 1.000 |       |            |

```

=== Homogeneity of dispersion (by color) ===
      Df    Sum Sq   Mean Sq  F value  Pr(>F)
Groups    1  0.00014  0.000145   0.0061  0.938
Residuals 130  3.07863  0.023682

```

```

=== dbRDA marginal tests ===
      Df  SumOfSqs      F    Pr(>F)
color      1      0.2421  1.425    0.193
intensity   4      1.1300  1.663    0.048  *
dist_km     1      1.2271  7.222 <0.001  ***
mean_phase  1      0.2509  1.477    0.184
pct_nonforest 1      0.6241  3.673    0.007  **
brightness_dark 1      0.5661  3.332    0.012  *
Residual   122     20.7295

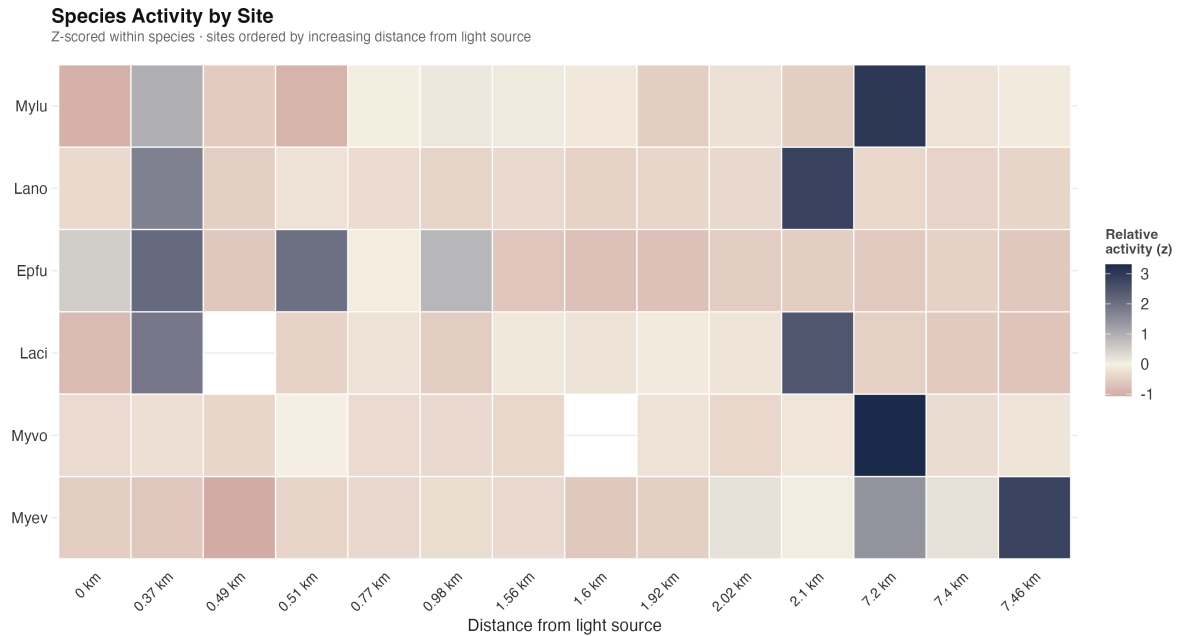
```

```

=== mvabund -- multivariate GLM (nBoot = 9999) ===
      Dev   Df  Pr(>Dev)
color      9.22   1    0.415
intensity  83.52   4    0.002  **
dist_km   142.57   1 <2e-16  ***
mean_phase  24.59   1    0.016  *
pct_nonforest 40.16   1    0.001  ***
brightness_d 54.29   1 <2e-16  ***

```

Color does not restructure community composition (dbRDA  $p = 0.193$ , mvabund  $p = 0.415$ ). Distance is the dominant structuring variable (dbRDA  $F = 7.22$ ,  $p < 0.001$ ; mvabund Dev = 142.6,  $p < 0.001$ ). Intensity marginally significant in both analyses. Habitat and sky brightness also contribute. Dispersion homogeneous between color treatments ( $p = 0.938$ ). PERMANOVA result reflects centroid location, not spread.



**Figure 3:** Species activity by site (z-scored within species), sites ordered by increasing distance from the Colter Bay parking lot.

## SPECIES-LEVEL RESULTS

=== mvabund univariate p-values (adjusted, PIT-trap 9999 iter) ===

| Species | Color | Intensity | Distance | Moon phase | %nonforest | brightness |
|---------|-------|-----------|----------|------------|------------|------------|
| Epfu    | ns    | ***       | **       | *          | ns         | ns         |
| Laci    | ns    | ns        | **       | ns         | ns         | ns         |
| Lano    | ns    | ns        | **       | ns         | ns         | ns         |
| Myev    | ns    | ns        | ***      | ns         | **         | ns         |
| Mylu    | ns    | ns        | *        | ns         | ns         | ns         |
| Myvo    | ns    | ns        | ***      | ns         | *          | ***        |
| Myyu    | ns    | ns        | ns       | ns         | ns         | ns         |
| Myci    | ns    | ns        | ns       | ns         | ns         | ns         |

Epfu: most treatment-sensitive; responds to intensity, distance, and moon phase. Myev: tracks habitat (pct\_nonforest) and distance. Myvo: strongly tracks sky brightness and distance. Color is non-significant for all species (Supplemental Figs. S8-S12).

## BAT MORPHOLOGY TRAITS

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We have four continuous wing-morphology measurements per species: ear height, forearm length, wing loading, and aspect ratio. Each trait was modeled separately (z-scored, interacted with intensity and distance) and compared at representative low / mean / high ( $\pm 1$  SD) values.

All four traits point the same direction. Species with smaller ears, shorter forearms, lower wing loading, and lower aspect ratio -- slower, more maneuverable fliers such as *Myotis* spp. -- show an increase in activity with distance from the Colter Bay parking lot, while species at the high end of each trait (larger, faster, open-air-adapted fliers such as *Lasi*) are comparatively flat across the same gradient. The same ordering holds for intensity: low-trait-value species have consistently higher predicted detections at every intensity level, with the gap narrowing at high intensity. Because forearm length, wing loading, ear height, and aspect ratio are all measured on the same six species, the four traits are highly correlated with one another (and with species identity itself, as guild was) -- the four panels are best read as one wing-morphology gradient viewed four ways, not four independent effects (Supplemental Figs. S13-S14).

## MOTH (LEPIDOPTERA) DATA

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### Data and General Activity Model

Moth specimens were identified from nightly traps at the same 14 monitoring stations, over the same 2022 season (21 June - 16 August). Each photo's filename encoded site, date, and light treatment (e.g. GRTE01\_2aug22\_w100D); intensity readings that deviated from the bat project's 10/30/50/70/100% schedule were rounded to the nearest level, and repeat IDs of the same site/date/treatment were merged. 9,864 identified specimens across 241 identifications (225 site-nights) were retained, spanning 11 families and 31 subfamilies, dominated numerically by Noctuidae, Lasiocampidae, and Erebididae.

A general activity model -- mirroring the bat Primary Model exactly (color  $\times$  intensity + intensity  $\times$  dist\_km + mean\_phase + pct\_nonforest + jd\_c + jd\_c<sup>2</sup> + (1|site), negative binomial, zero-inflated) -- was fit to total nightly moth counts.

```

Family:      nbinom2  ( log )
Formula:     detections ~ color * intensity + intensity * dist_km +
              mean_phase + pct_nonforest + jd_c + I(jd_c^2) + (1 | site)
Zero infl:   ~1
Data:        insect_env

```

| AIC    | BIC    | logLik | -2*log(L) | df.resid |
|--------|--------|--------|-----------|----------|
| 1992.0 | 2067.2 | -974.0 | 1948.0    | 204      |

Random effects:

| Groups | Name        | Variance | Std.Dev. |
|--------|-------------|----------|----------|
| site   | (Intercept) | 0.0122   | 0.1103   |

Number of obs: 226, groups: site, 14

Dispersion parameter for nbinom2 family (): 2.43

Conditional model:

|                      | Estimate | Std.Err | z value | Pr(> z ) |     |
|----------------------|----------|---------|---------|----------|-----|
| (Intercept)          | 4.1115   | 0.2755  | 14.924  | <2e-16   | *** |
| colorW               | 0.0596   | 0.2689  | 0.222   | 0.8244   |     |
| intensity30          | 0.3922   | 0.2996  | 1.309   | 0.1905   |     |
| intensity50          | 0.0481   | 0.2987  | 0.161   | 0.8722   |     |
| intensity70          | 0.1013   | 0.2827  | 0.358   | 0.7201   |     |
| intensity100         | -0.6289  | 0.2818  | -2.232  | 0.0256   | *   |
| dist_km              | 0.1086   | 0.0408  | 2.661   | 0.0078   | **  |
| mean_phase           | -0.4558  | 0.2414  | -1.888  | 0.0590   | .   |
| pct_nonforest        | -0.2558  | 0.2164  | -1.182  | 0.2371   |     |
| jd_c                 | 0.0056   | 0.0036  | 1.528   | 0.1266   |     |
| I(jd_c^2)            | -0.0028  | 0.0003  | -9.562  | <2e-16   | *** |
| colorW:intensity30   | -0.3184  | 0.3806  | -0.837  | 0.4028   |     |
| colorW:intensity50   | -0.0686  | 0.3578  | -0.192  | 0.8479   |     |
| colorW:intensity70   | 0.1072   | 0.3414  | 0.314   | 0.7535   |     |
| colorW:intensity100  | 1.0669   | 0.3252  | 3.280   | 0.0010   | *** |
| intensity30:dist_km  | 0.0490   | 0.0579  | 0.846   | 0.3978   |     |
| intensity50:dist_km  | 0.0393   | 0.0547  | 0.718   | 0.4727   |     |
| intensity70:dist_km  | 0.0264   | 0.0544  | 0.484   | 0.6282   |     |
| intensity100:dist_km | -0.1158  | 0.0523  | -2.214  | 0.0268   | *   |

Unlike bats, moths show no significant color preference at low-to-mid intensity.

At 100% intensity, however, white light produced 3.1× more detections than red ( $p < 0.0001$ ) -- the mirror image of the bat pattern at that same intensity (Section 2), where red and white were statistically indistinguishable. The positive distance slope seen at 10-70% intensity flattens to essentially zero at 100%, echoing the bat finding that the distance gradient weakens at high intensity (Supplemental Figs. S14-S15).

### Family-Level Models

Six of nine families cleared the same 30-detection / 20-row threshold used for the bat species models (Noctuidae, Lasiocampidae, Erebidae, Geometridae, Sphingidae, Notodontidae); Crambidae, Drepanidae, and Euteliidae had too few identified specimens to model individually. Noctuidae is numerically dominant (predicted 40-50 detections/night vs. <15 for every other family) and drives most of the aggregate pattern above. At 100% intensity, Erebidae, Noctuidae, and Geometridae each show a 2-3× white-over-red ratio; at lower intensities the families track each other closely with wide, overlapping confidence intervals (Supplemental Figs. S16-S17).

### Community Analysis

=== PERMANOVA (adonis2), family-level community ===

|          | Df  | SumOfSqs | R2    | F     | Pr(>F) |     |
|----------|-----|----------|-------|-------|--------|-----|
| Model    | 9   | 6.781    | 0.305 | 5.943 | <0.001 | *** |
| Residual | 122 | 15.468   | 0.695 |       |        |     |
| Total    | 131 | 22.249   | 1.000 |       |        |     |

=== dbRDA marginal tests ===

|                 | Df  | SumOfSqs | F     | Pr(>F) |     |
|-----------------|-----|----------|-------|--------|-----|
| color           | 1   | 0.201    | 1.581 | 0.159  |     |
| intensity       | 4   | 3.930    | 7.749 | <0.001 | *** |
| dist_km         | 1   | 1.065    | 8.401 | <0.001 | *** |
| mean_phase      | 1   | 0.761    | 6.005 | 0.001  | **  |
| pct_nonforest   | 1   | 0.104    | 0.821 | 0.494  |     |
| brightness_dark | 1   | 0.086    | 0.682 | 0.596  |     |
| Residual        | 122 | 15.468   |       |        |     |

=== mvabund --- multivariate GLM (nBoot = 9999) ===

|       | Dev  | Df | Pr(>Dev) |
|-------|------|----|----------|
| color | 6.11 | 1  | 0.585    |

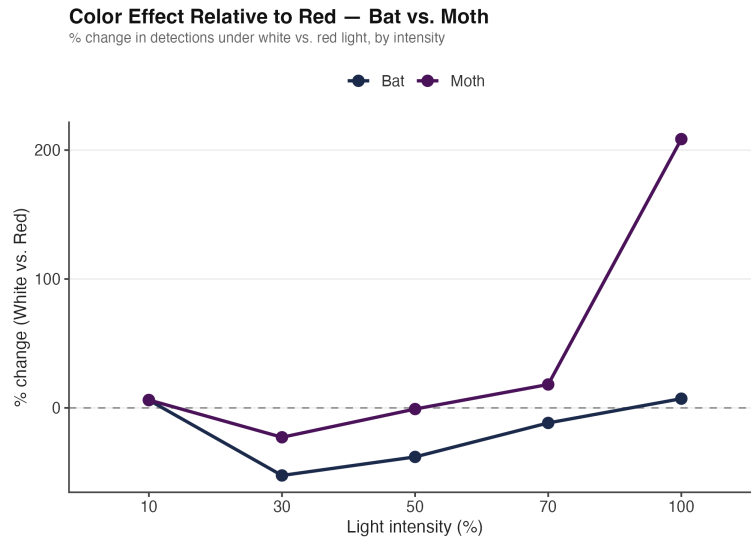
|               |        |   |        |     |
|---------------|--------|---|--------|-----|
| intensity     | 173.32 | 4 | <2e-16 | *** |
| dist_km       | 73.77  | 1 | <2e-16 | *** |
| mean_phase    | 40.72  | 1 | <2e-16 | *** |
| pct_nonforest | 13.34  | 1 | 0.149  |     |
| brightness_d  | 6.82   | 1 | 0.500  |     |

As with bats, light *color* has no detectable effect on moth community composition (dbRDA  $p = 0.159$ , mvabund  $p = 0.585$ ). Intensity, distance, and moon phase all structure the community strongly (dbRDA  $p < 0.001$  for all three; mvabund  $p < 2e-16$ ). Notably, the treatment variables jointly explain far more of the moth community's variation (model  $R^2 = 0.305$ ) than the bat community's (model  $R^2 = 0.156$ , Community Analysis section above) -- moth community turnover along the light gradient is proportionally stronger than bats'. Per-family univariate tests attribute most of the intensity signal to Lasiocampidae and Noctuidae (both  $p < 2e-16$ ) and the distance signal primarily to Lasiocampidae ( $p < 2e-16$ ) and Noctuidae ( $p = 0.005$ ) (Supplemental Figs. S18-S19).

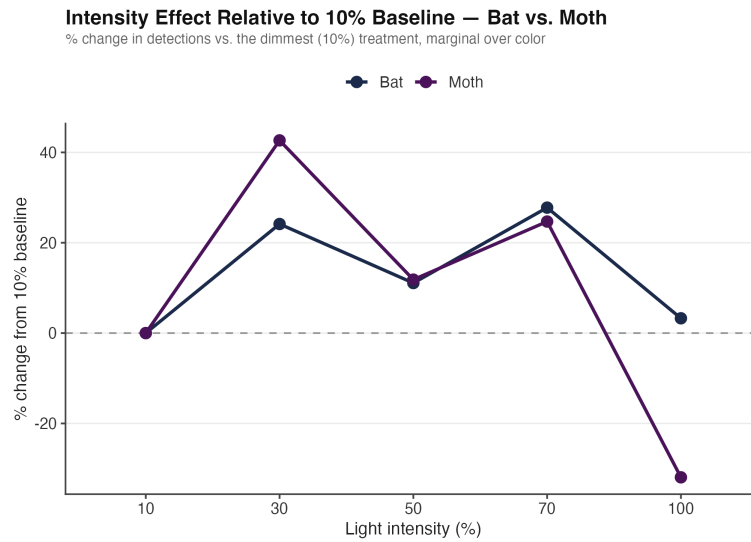
#### BAT VS. MOTH COMPARISON

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Raw totals differ by roughly an order of magnitude between taxa and reflect different sampling effort rather than treatment response (Supplemental Figs. S20-S22). The figures below instead rescale each model's marginal predictions to % change from that taxon's own baseline, comparing the *shape* of each taxon's response independent of its absolute abundance.

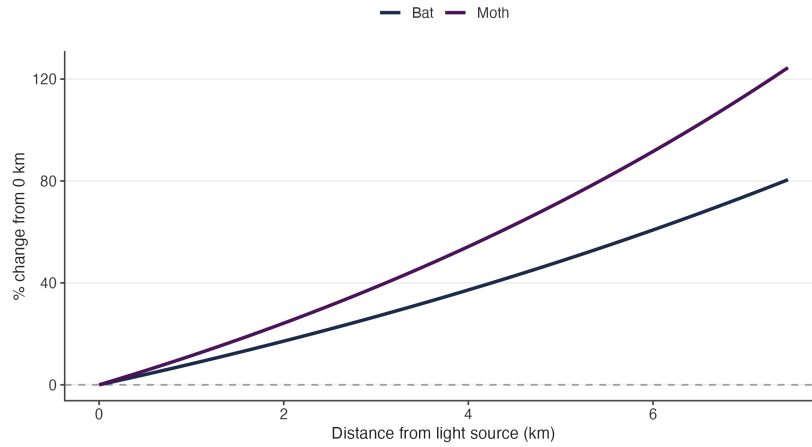


**Figure 4:** % change in detections, white vs. red light, by intensity -- both taxa on a shared scale.



**Figure 5:** % change in detections from the 10% baseline, marginal over color -- both taxa on a shared scale.

**Distance Gradient Relative to the Nearest Shared Site — Bat vs. Moth**  
 % change in detections vs. 0 km - marginal over color and intensity



**Figure 6:** % change in detections along the distance gradient from the nearest shared site -- both taxa on a shared scale.

A companion effect-size forest plot (rate ratios  $\pm$  95% CI for color, each intensity level vs. 10%, and the marginal distance slope) is provided in Supplemental Fig. S23; the same estimates are summarized in the table below.

| Term                  | Bat ratio [95% CI] |              | Moth ratio [95% CI] |              |
|-----------------------|--------------------|--------------|---------------------|--------------|
| Color (White vs Red)  | 0.78               | [0.71, 0.86] | 1.24                | [1.02, 1.51] |
| Intensity 30% vs 10%  | 1.24               | [0.99, 1.55] | 1.43                | [0.96, 2.11] |
| Intensity 50% vs 10%  | 1.11               | [0.86, 1.44] | 1.12                | [0.71, 1.77] |
| Intensity 70% vs 10%  | 1.28               | [1.11, 1.47] | 1.25                | [0.92, 1.69] |
| Intensity 100% vs 10% | 1.03               | [0.87, 1.22] | 0.68                | [0.48, 0.97] |
| Distance (per km)     | 1.08               | [0.95, 1.23] | 1.11                | [1.07, 1.16] |

The two taxa diverge most clearly on light color: averaged across intensity, bats show 22% fewer detections under white than red (CI excludes 1), while moths show 24% more (CI excludes 1) -- opposite directions. At 100% intensity specifically, bat activity is indistinguishable from the 10% baseline (ratio 1.03, ns) while moth activity drops significantly (ratio 0.68, CI excludes 1): bats hold steady at maximum brightness where moths pull back. Both taxa increase with distance from the Colter Bay parking lot over the shared 0–7.4 km range, and the relative increase is if anything larger for moths (+125% at the far end vs. +80% for bats); only the moth slope is significant on its own (CI excludes 1), because the bat's marginal, intensity-averaged slope is diluted by the reversal at 70–100% intensity documented in the Primary Model (Section 2) -- a reversal not present in the moth data.

## SUMMARY

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1. Color × intensity interaction: red light at 30–50% produced 1.6–2.1× more bat detections than white light. At 70% and 100%, colors converge. White light is flat across all intensities.
2. Intensity × distance interaction: 70% and 100% intensities show significant negative distance slopes (intensity70:dist\_km  $\beta = -0.128$ ,  $p < 0.001$ ; intensity100  $\beta = -0.090$ ,  $p = 0.001$ ), meaning bat detections under high light intensity increase with distance from Colter Bay parking lot. 30% and 50% slopes are non-significant -- low light intensities are distance-invariant.
3. Community composition is driven by distance, habitat (pct\_nonforest), and sky brightness, not by light color.
4. Wing morphology (ear height, forearm length, wing loading, aspect ratio): smaller, slower, more maneuverable species consistently show higher and more distance-sensitive activity than larger, faster, open-air fliers, across every intensity level.
5. Moths mirror bats on intensity and distance (both show a positive distance effect that weakens at 100% intensity, and light color has no effect on community composition in either taxon) but invert the color effect: bats favor red overall while moths favor white, most sharply at 100% intensity (bat ratio 1.03 ns vs. moth ratio 0.68, CI excludes 1).
6. Red light at 30–50% intensity minimises suppression of bat activity while also avoiding the white-light attraction effect seen in moths at high intensity; siting lights >0.5–1.0 km from sensitive habitat remains important for both taxa at high intensity.

## SPECIES KEY

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|      |                                  |                               |
|------|----------------------------------|-------------------------------|
| Epfu | <i>Eptesicus fuscus</i>          | (big brown bat)               |
| Laci | <i>Lasiurus cinereus</i>         | (hoary bat)                   |
| Lano | <i>Lasionycteris noctivagans</i> | (silver-haired bat)           |
| Myci | <i>Myotis ciliolabrum</i>        | (western small-footed myotis) |
| Myev | <i>Myotis evotis</i>             | (long-eared myotis)           |
| Mylu | <i>Myotis lucifugus</i>          | (little brown myotis)         |
| Myvo | <i>Myotis volans</i>             | (long-legged myotis)          |

Myyu Myotis yumanensis (Yuma myotis)

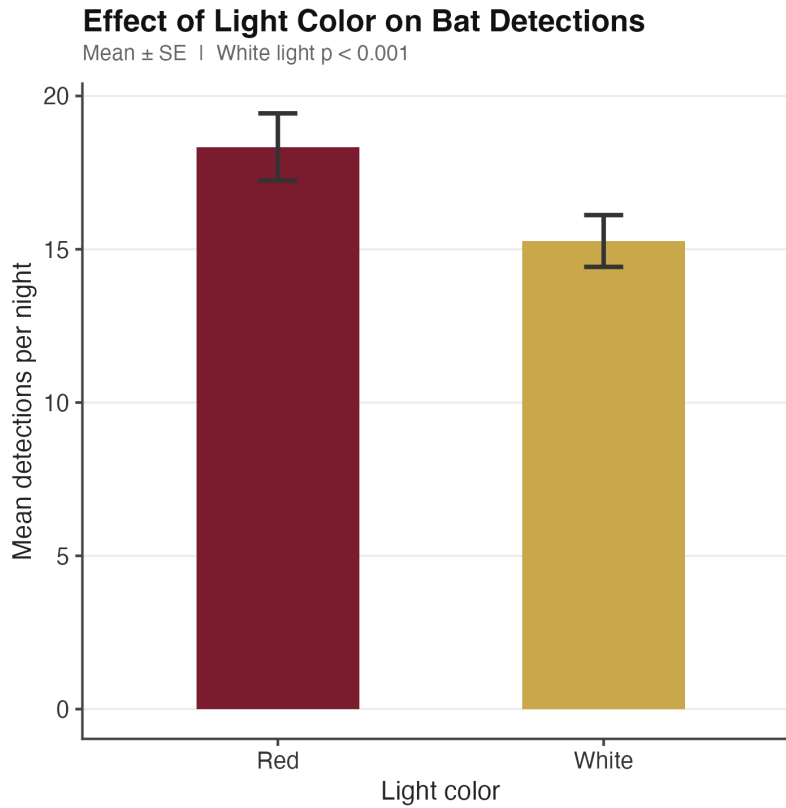
Moth families modeled individually (Section 9):

|               |                   |                              |
|---------------|-------------------|------------------------------|
| Noctuidae     | Owlet moths       | (numerically dominant)       |
| Lasiocampidae | Lappet/tent moths |                              |
| Erebidae      | Erebid moths      | (incl. tiger, tussock moths) |
| Geometridae   | Geometer moths    | ("inchworm" moths)           |
| Sphingidae    | Sphinx/hawk moths |                              |
| Notodontidae  | Prominent moths   |                              |

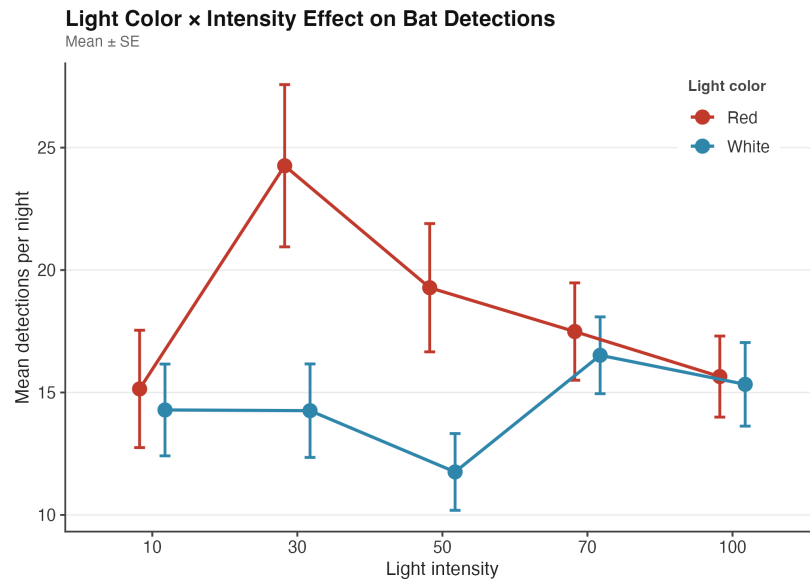
## SUPPLEMENTAL MATERIALS

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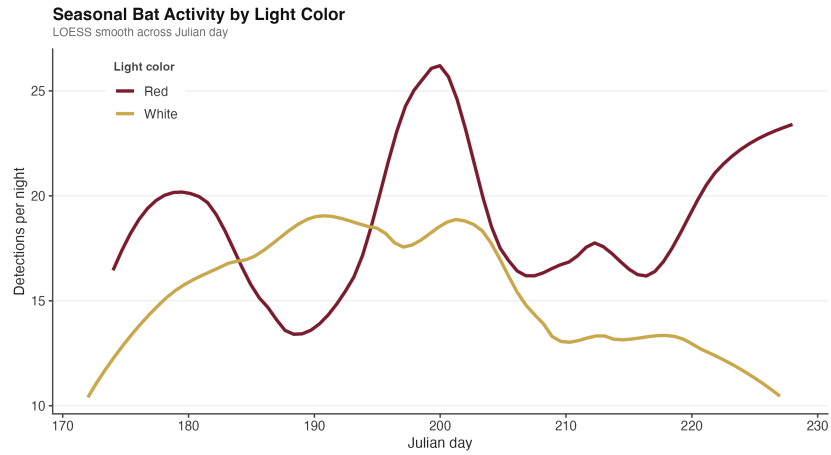
Additional figures supporting the results in the main text, in the order those results are discussed above.



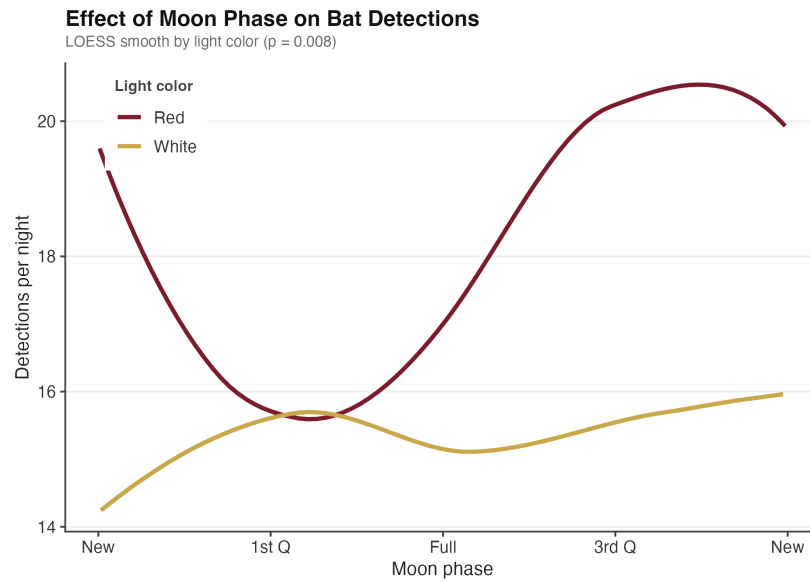
**Figure S1:** Mean detections per night by light color (cardinal crimson = red, gilded gold = white), averaged across intensities. Error bars = SE.



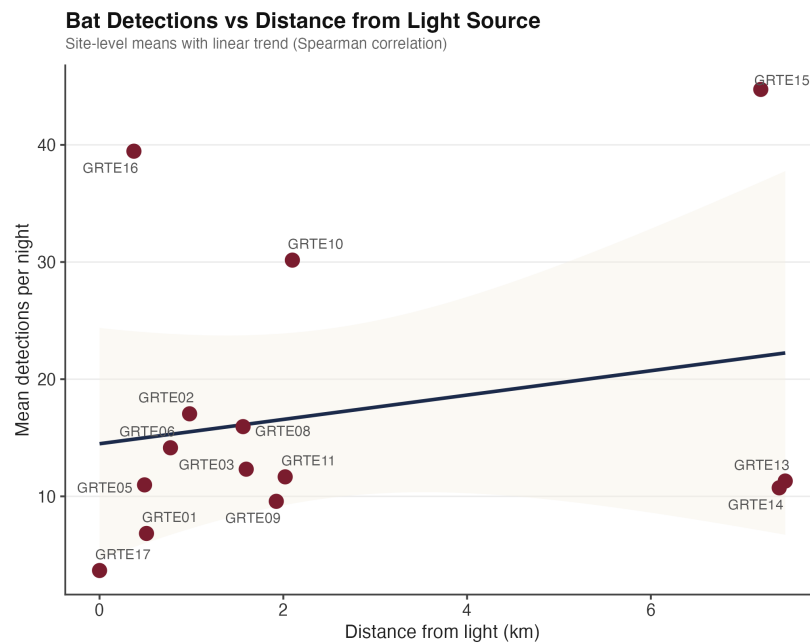
**Figure S2:** Mean detections per night by color at each intensity level with 95% CI. Cardinal crimson (red) peaks at 30% and converges with gilded gold (white) at high intensity.



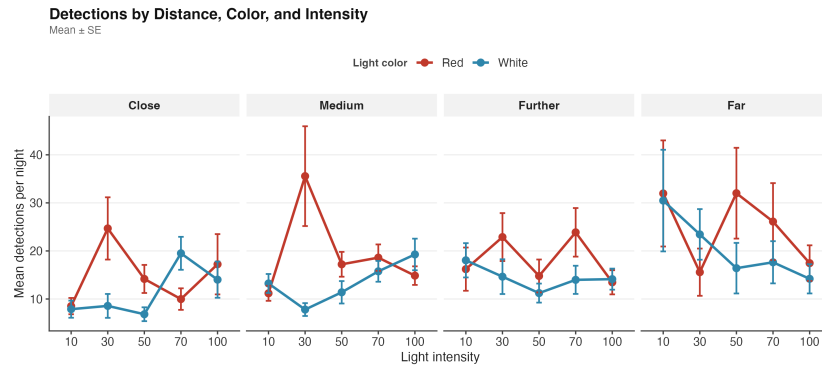
**Figure S3:** Seasonal trend in bat detections by light color. LOESS smoothed. Cardinal crimson (red) and gilded gold (white) both decline through the season; red/white separation persists across the survey period.



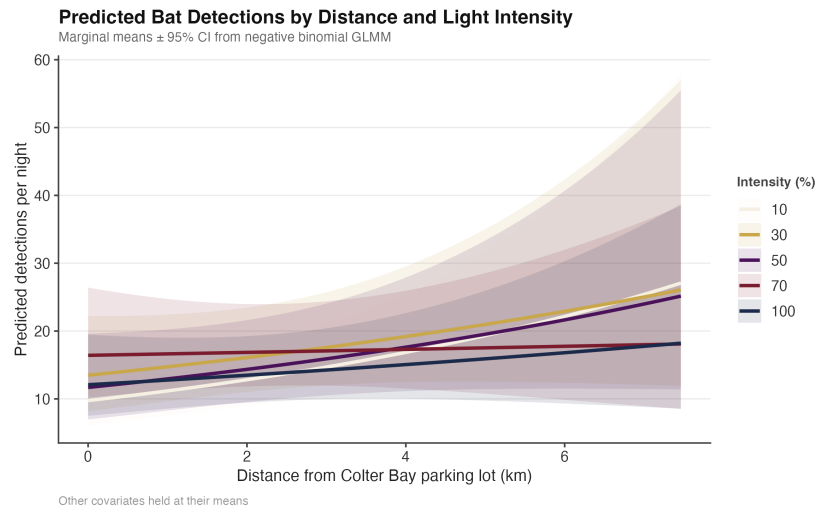
**Figure S4:** Bat detections vs. moon phase by color (cardinal crimson = red, gilded gold = white). No evident color  $\times$  moon interaction.



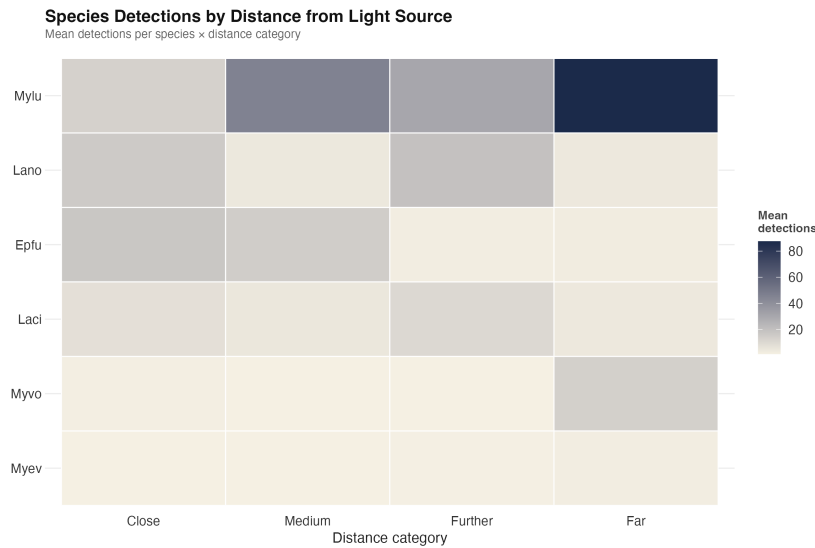
**Figure S5:** Site-level mean detections vs. distance from Colter Bay parking lot (km). Points in cardinal crimson; navy trend line with manuscript ivory confidence band.



**Figure S6:** Detections by distance category and light treatment.



**Figure S7:** Predicted bat detections across the distance gradient by light intensity level. Marginal means  $\pm$  95% CI from the negative binomial GLMM. Intensity palette: manuscript ivory (10%), gilded gold (30%), papal purple (50%), cardinal crimson (70%), nave navy (100%). Steeper recovery with distance at 70% and 100% reflects the significant intensity  $\times$  distance interaction.

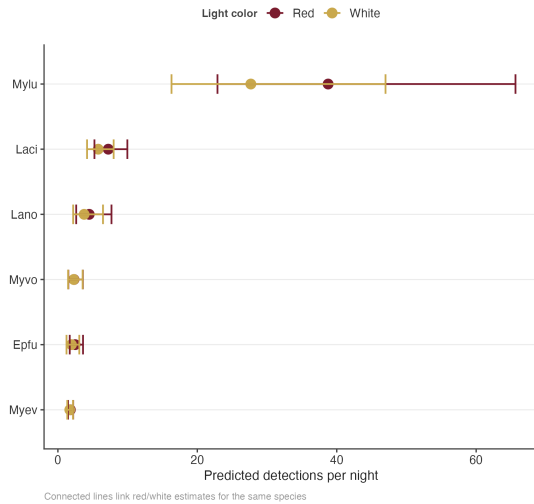


**Figure S8:** Species activity by distance category (z-scored within species). Scale runs manuscript ivory (low) to navae navy (high).

#### Species-Level Light Effects — Distance Project

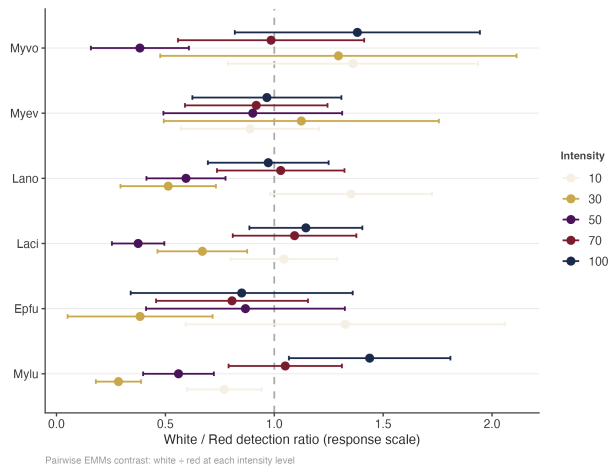
##### Light Color Effect on Bat Detections by Species

Marginal means  $\pm$  95% CI from per-species negative binomial GLMMs

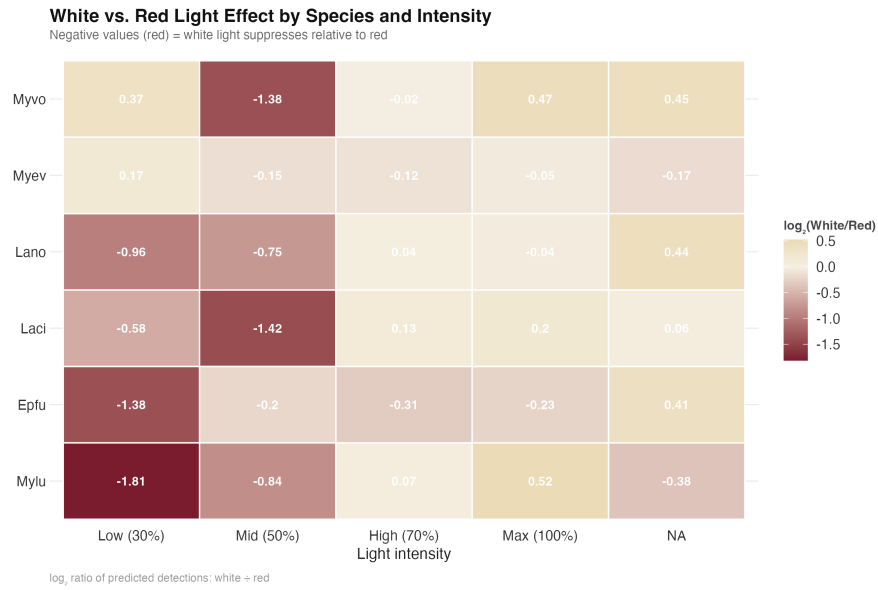


##### White vs. Red Light — Detection Ratio by Species & Intensity

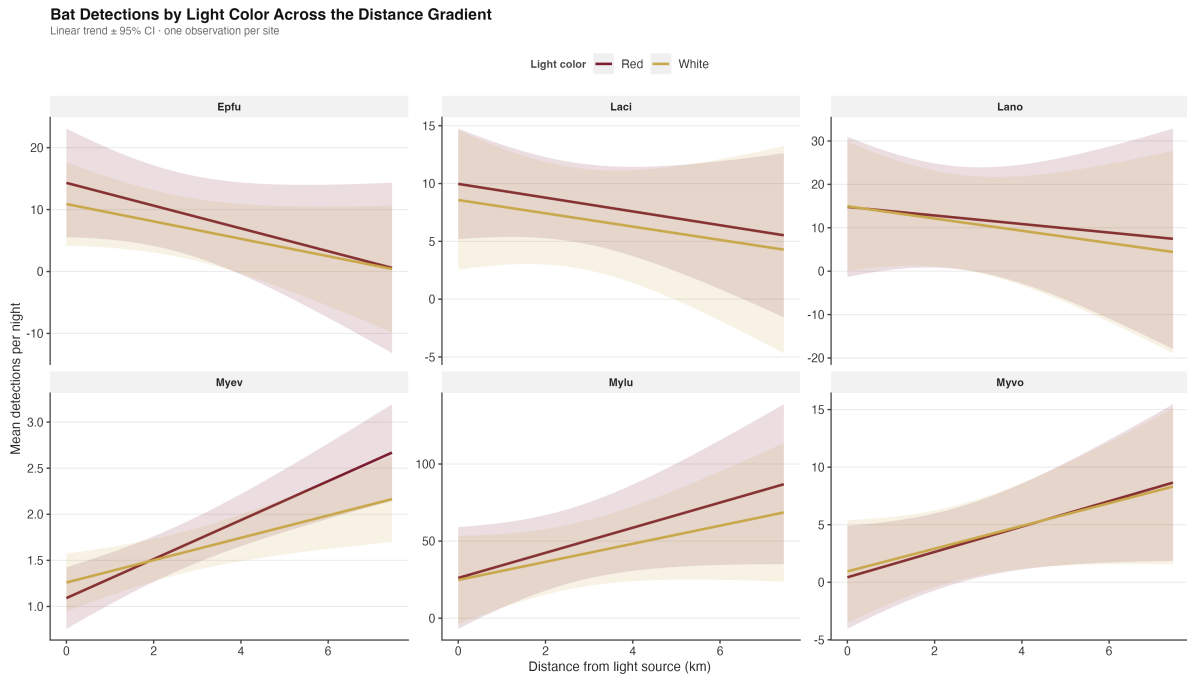
Ratio > 1 = more detections under white; < 1 = more under red



**Figure S9:** Left: marginal means  $\pm$  95% CI by color (cardinal crimson = red, gilded gold = white), ranked by species. Right: white/red detection ratio by intensity level per species (Catholic intensity palette). Ratio > 1 = more detections under white; < 1 = more under red.



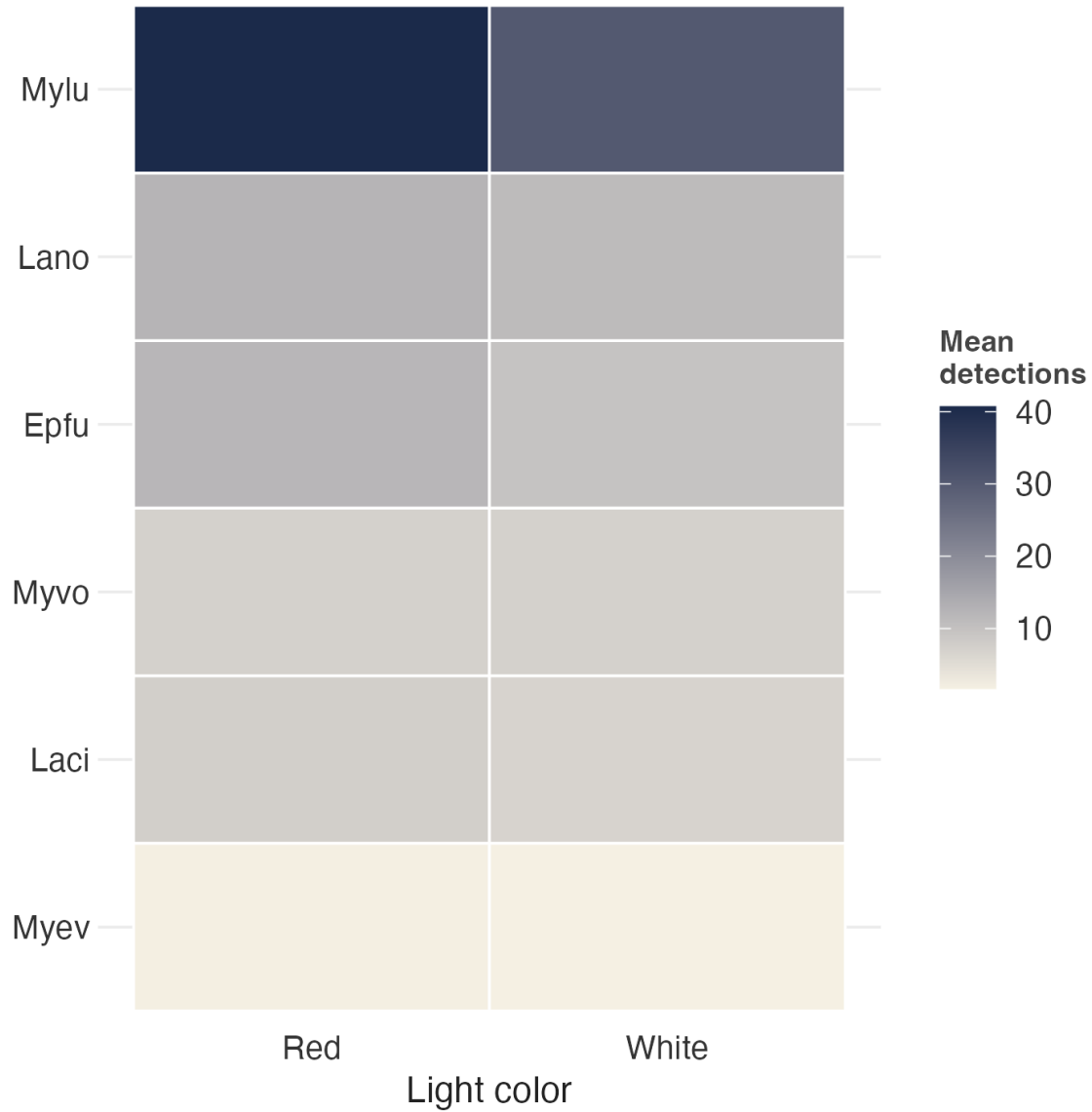
**Figure S10:** Log<sub>2</sub>(white/red) heatmap by species and intensity. Negative values (cardinal crimson) = white suppresses relative to red; positive values (gilded gold) = white enhances. Manuscript ivory = parity.



**Figure S11:** Bat detections by light color across the distance gradient, faceted by species. Linear trend ± 95% CI; one observation per site. Cardinal crimson = red, gilded gold = white.

## Bat Community Response by Light Color

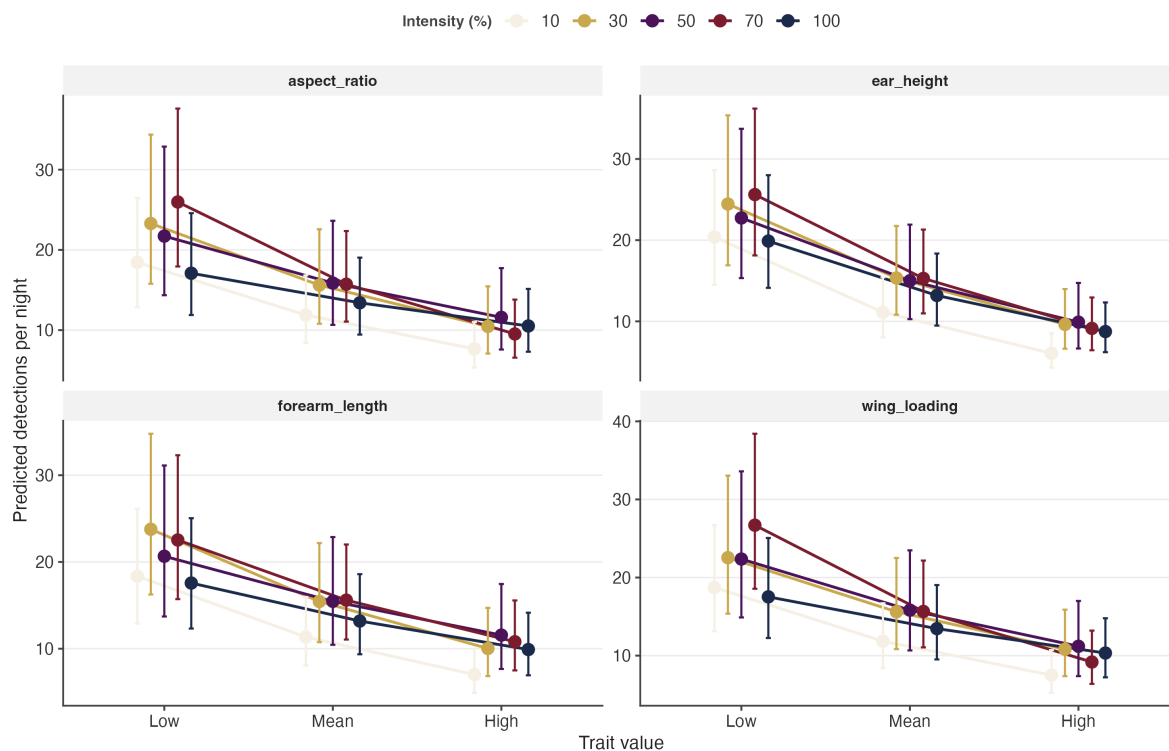
Mean detections per species



**Figure S12:** Species activity heatmap (z-scored within species) by light color and intensity. Scale runs manuscript ivory (low) to navy (high).

### Light Intensity Effect Across Morphology Trait Values

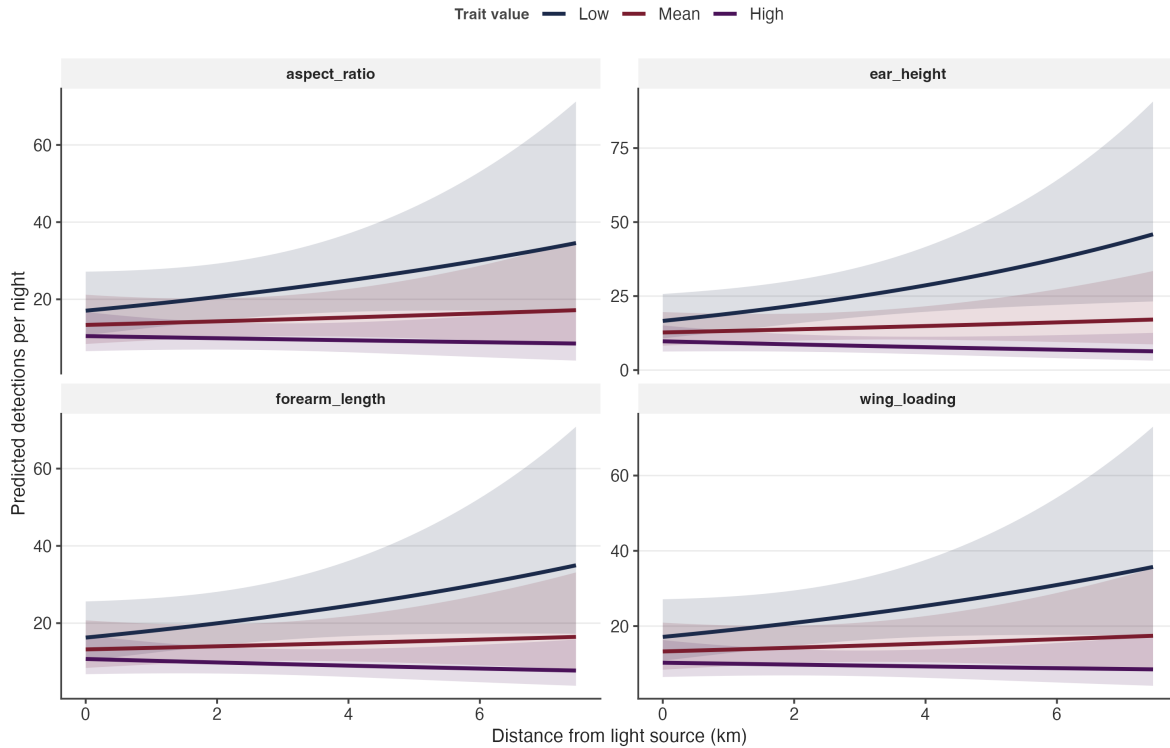
Marginal means  $\pm$  95% CI at low/mean/high ( $\pm 1$  SD) trait values



**Figure S13:** Predicted bat detections by light intensity, evaluated at low/mean/high ( $\pm 1$  SD) values of each morphology trait.

### Bat Activity Across Distance Gradient by Morphology Trait Value

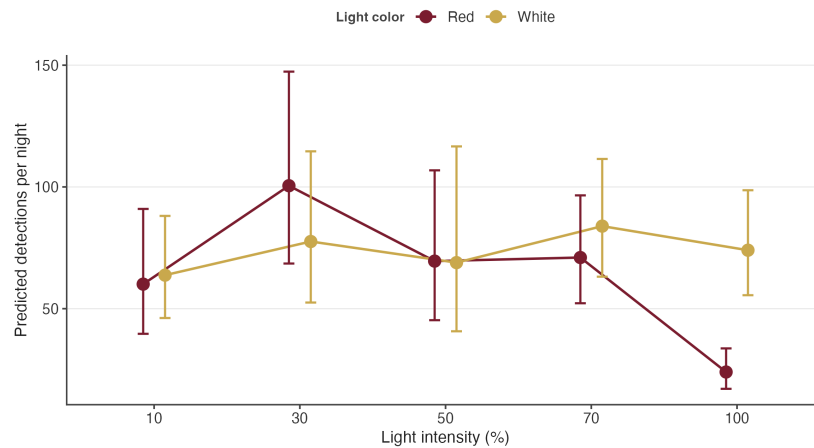
Marginal predictions  $\pm$  95% CI · low/mean/high ( $\pm 1$  SD) trait values



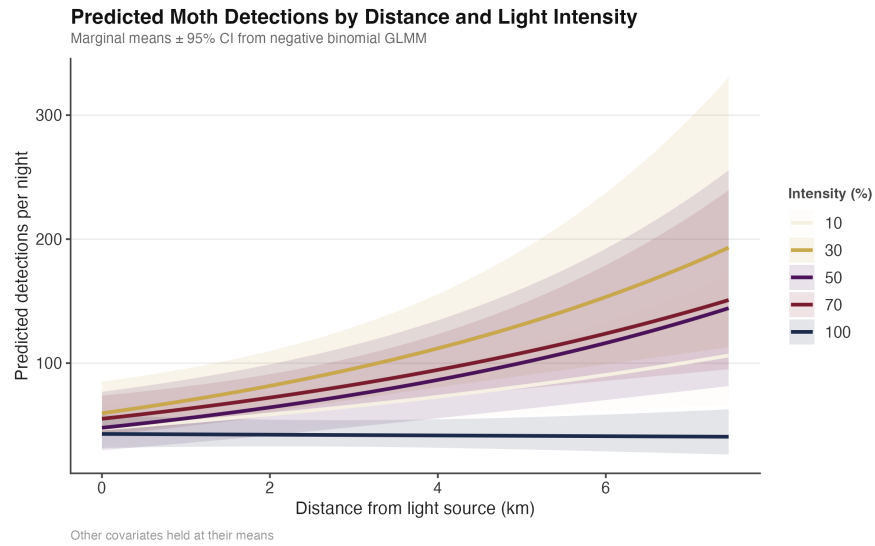
**Figure S14:** Predicted bat detections across the distance gradient, evaluated at low/mean/high ( $\pm 1$  SD) values of each morphology trait.

### Light Color $\times$ Intensity Effect on Moth Detections

Marginal means  $\pm$  95% CI from negative binomial GLMM

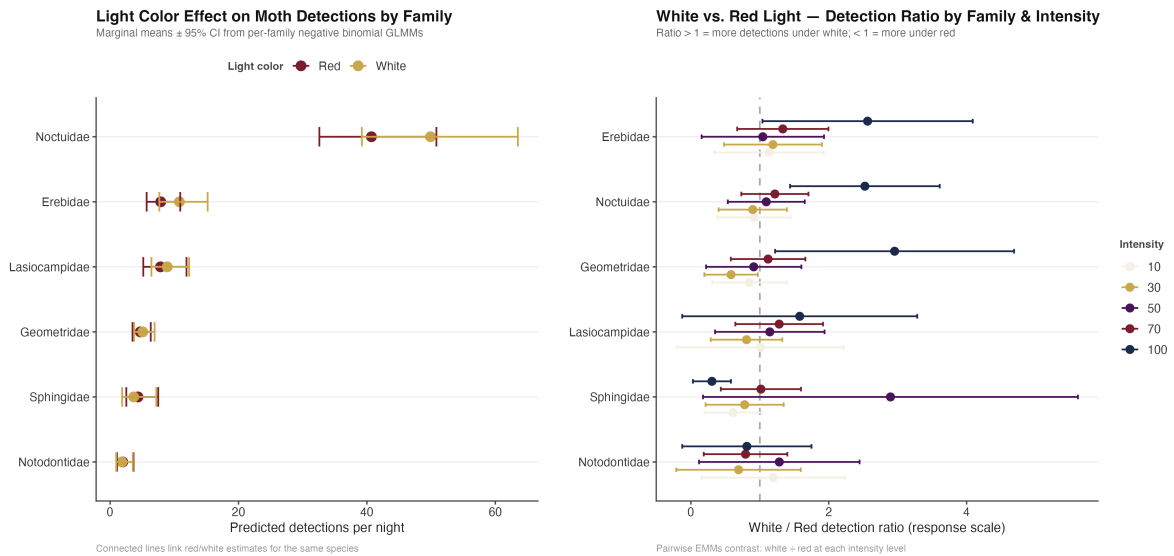


**Figure S15:** Moth detections by light color and intensity (general activity model). Cardinal crimson = red, gilded gold = white.

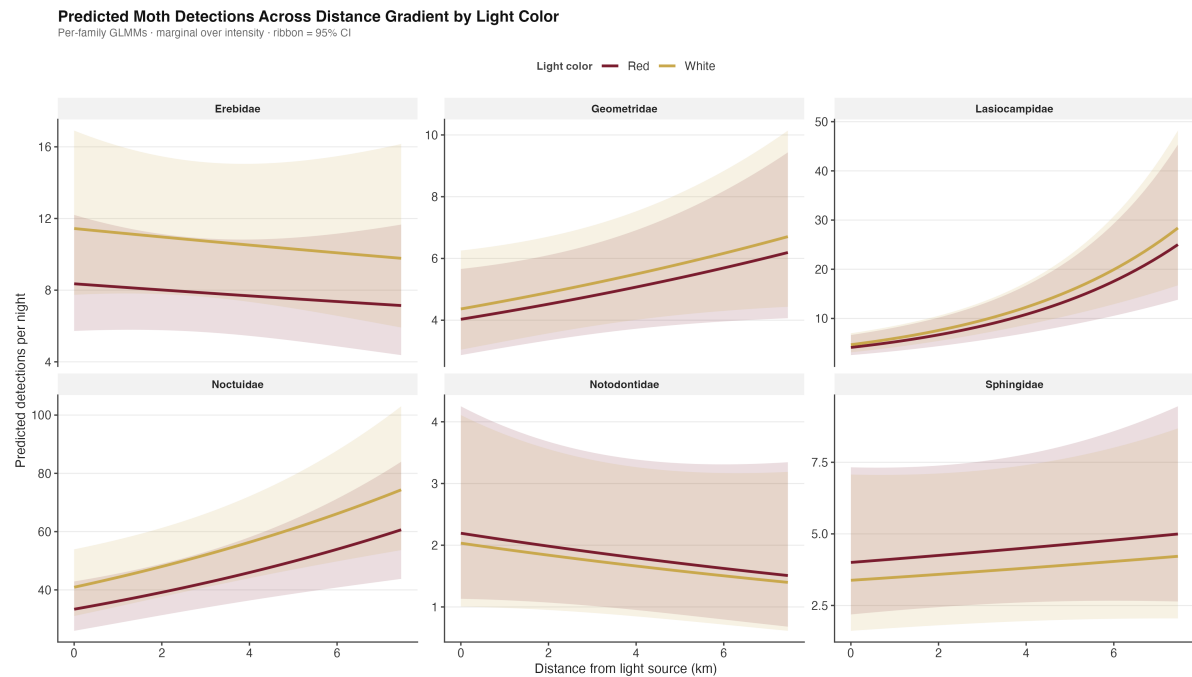


**Figure S16:** Predicted moth detections by distance and light intensity.

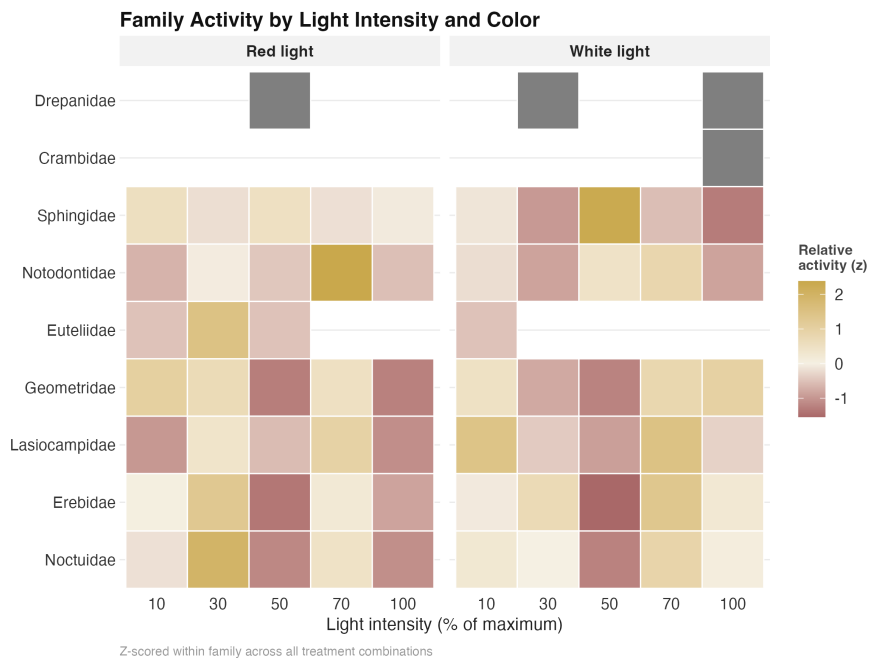
#### Family-Level Light Effects — Distance Project



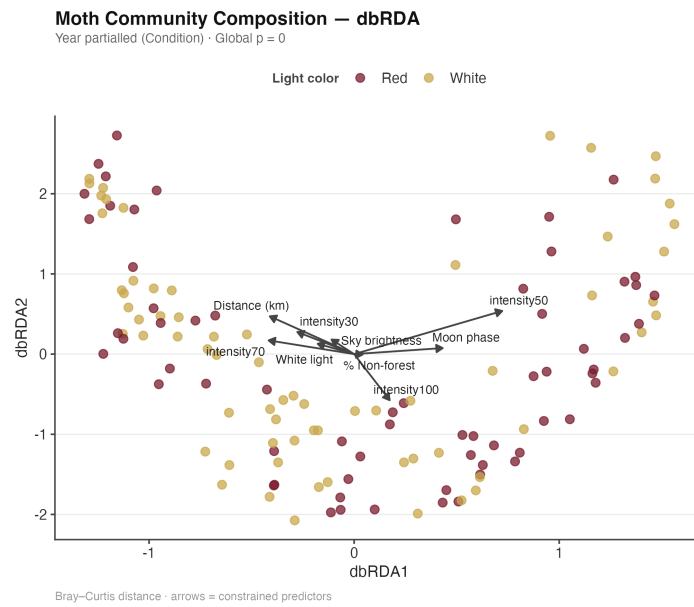
**Figure S17:** Left: marginal means  $\pm$  95% CI by color, ranked by family. Right: white/red detection ratio by intensity level per family. Ratio > 1 = more detections under white; < 1 = more under red.



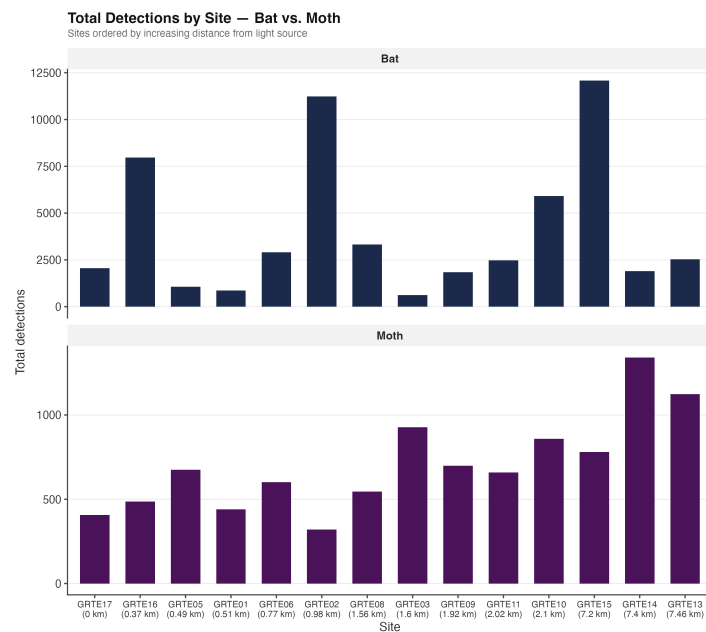
**Figure S18:** Predicted moth detections across the distance gradient by light color, faceted by family.



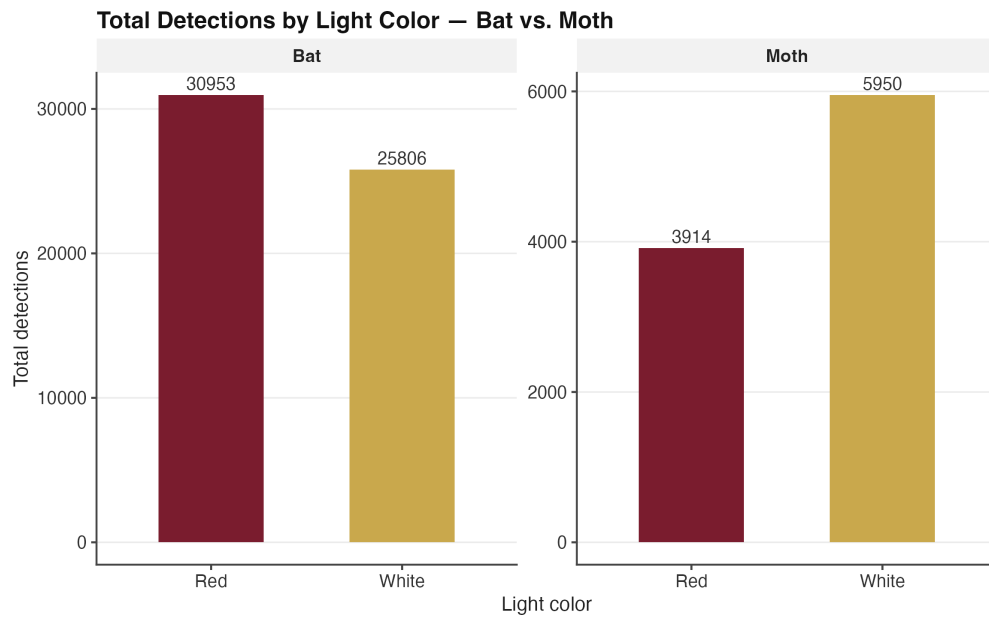
**Figure S19:** Moth family activity by light intensity and color (z-scored within family).



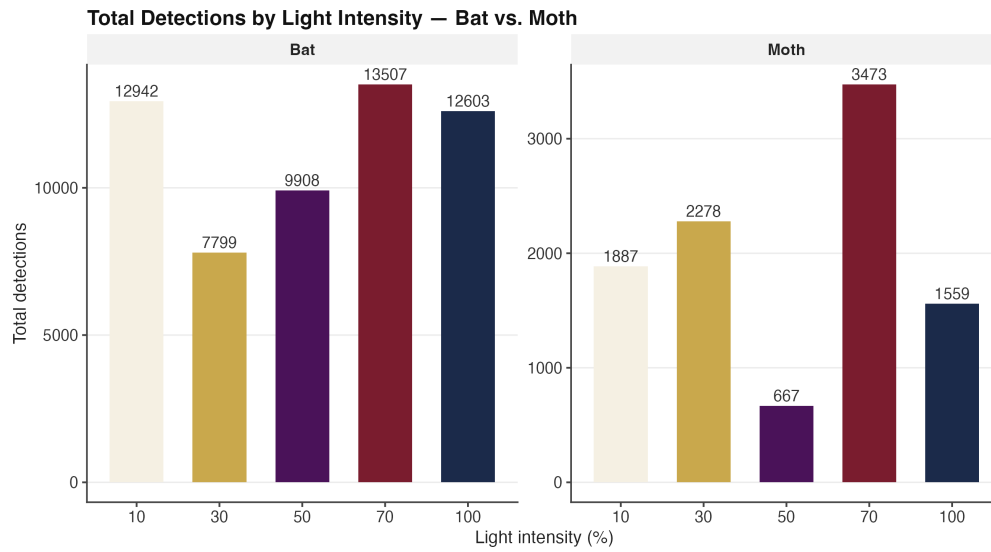
**Figure S20:** Moth family community composition -- dbRDA ordination.



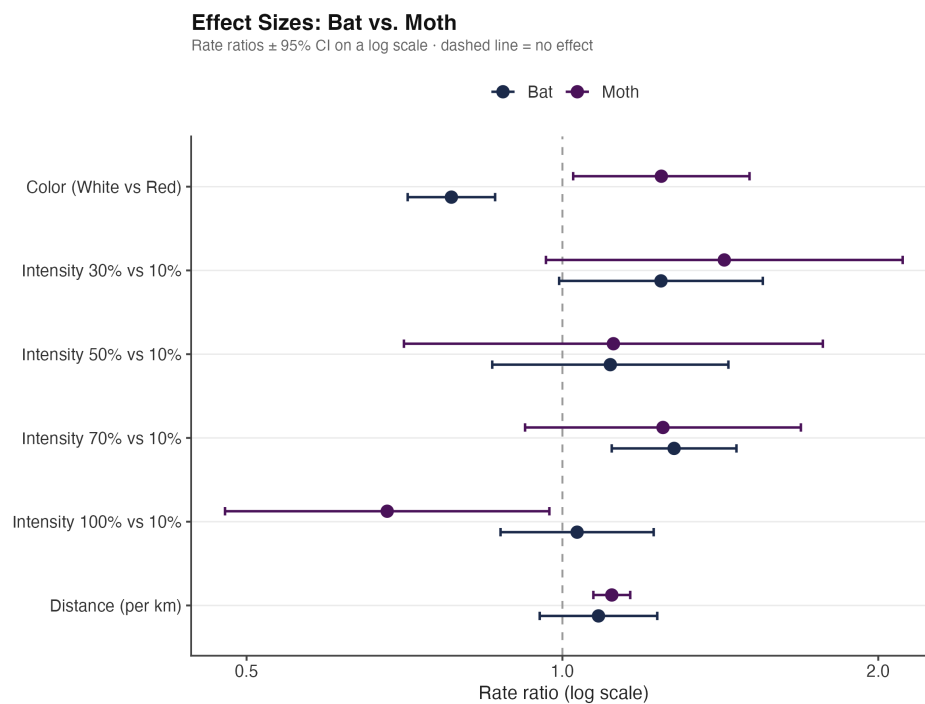
**Figure S21:** Total detections by site, bat vs. moth (sites ordered by distance from the Colter Bay parking lot). Bat = nave navy, Moth = papal purple.



**Figure S22:** Total detections by light color, bat vs. moth.



**Figure S23:** Total detections by light intensity, bat vs. moth.



**Figure S24:** Effect-size comparison: rate ratios  $\pm$  95% CI for color, each intensity level vs. 10%, and the marginal distance slope.